Econometrics I

Lecture 11b: Synthetic Control

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Fundamental Problem of Causal Inference

- We don't observe the counterfactual $Y_i(1 - D_i)$.
- For a single individual we either observe $Y_i(1)$ or $Y_i(0)$ but never both!
  - ex: We don't see what your wage would have been if you didn't attend college.
  - ex: We might know your cholesterol before you took Lipitor, but we don't know what it would be today if you didn't take Lipitor.

$$Y_i = D_i \cdot Y_i(1) + (1 - D_i) \cdot Y_i(0)$$

i	$Y_i(1)$	$Y_i(0)$	D_i	X_i
1	1	?	1	3.2
2	0	?	1	1.7
3	?	0	0	2.5
		⋮		
n	?	1	0	-1.5

How do we construct the counterfactual?

We observe $Y_i(D_i)$ but not $Y_i(1 - D_i)$, how do we estimate it?

Event Study: Compare $Y_{t+k}(1)$ to $Y_{t-j}(0)$ before and after an event at t.

Matching: Compare $Y_i(1)$ to $Y_j(0)$ for $i \neq j$ but where $X_i = X_j$.

Difference in Difference: Compare $Y_{i,t}(1) - Y_{i,t-1}(0)$ to $Y_{j,t}(0) - Y_{j,t-1}(0)$ for some $j \neq i$.

Synthetic Control: Compare $Y_{i,t}(1)$ to $\sum_{j \neq i} w_j Y_{j,t}(0)$, a weighted average of other units with weights chosen so that the pre-period Y's and X's of i are matched.

Synthetic Control

Motivation: Recap

Difference in Difference approaches have some drawbacks:

- We need to really believe parallel trends
  - Is ΔPA really a good counterfactual for ΔNJ?
  - Obvious question: why not pick ΔDE or ΔNY?
  - Measured effect shouldn't change if our assumption is valid (but it probably will!)
- With multiple treated individuals we can use 2WFE estimator

$$y_{it} = \beta X_{it} + \delta_i \cdot D_{it} + \gamma_i + \gamma_t + u_{it}$$

  - It assumes additivity of FE $\gamma_i + \gamma_t$ are independent!
  - States have different baseline levels γ_i but evolution over time is common γ_t.
  - Turns out it doesn't really generalize the diff-in-diff with $I > 2$ (Imai and Kim 2021).

Motivation: Recap

Matching Estimators had drawbacks too:

- We need to believe in as if random-assignment conditional on X
- CIA: $\{Y_i(1), Y_i(0)\} \perp D_i | X_i$.

But we could use pretty flexible methods in constructing our matched controls:

- We could try to match on multiple dimensions of X.
- k–NN, kernels, etc.

What if we could use ideas from matching to better satisfy something like our parallel trends assumptions?

Example: Abadie, Diamond, Hainmueller (2010)

The Question

In 1988 California passes anti-smoking Prop 99

- increased excise tax by 25 cents per pack,
- earmarked the tax revenues to health and anti-smoking education budgets and funded anti-smoking ads
- led to indoor smoking bans in restaurants and bars city by city

What was the effect on per capita cigarette sales?

- Already a bunch of pre-existing trends.
- What is a good control for California?

Use state-level data from 1970-2000.

The Idea

Use a convex combination of other states to construct a synthetic counterfactual California.

- We observe Y_{it}, X_{it}, D_{it}.
- Assume only $i = 1$ and $t > T_0$ are treated.
- Construct a donor pool of potential controls subscripted by j.
- Choose some weights w_j for each entity (state) in donor pool. How?
  - Same $\mathbf{X_1} = \sum_j w_j \mathbf{X_j}$ as treated observations (like matching).
  - Same $\left(Y_{1,1}, \dots, Y_{1,T_0}\right) = \sum_j w_j \cdot \left(Y_{j,1}, \dots, Y_{j,T_0}\right)$ (like parallel trends).
  - Weights sum to one $\sum_j w_j = 1$ and maybe are non-negative (or not!)
- Idea is to match all of the X's and all of the Y_{it}'s in the pre-period

Table 1. Cigarette sales predictor means

Variables	California		Average of 38 control states
	Real	Synthetic	
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15–24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

NOTE: All variables except lagged cigarette sales are averaged for the 1980–1988 period (beer consumption is averaged 1984–1988). GDP per capita is measured in 1997 dollars, retail prices are measured in cents, beer consumption is measured in gallons, and cigarette sales are measured in packs.

Donor Weights

Table 2. State weights in the synthetic California

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Trend Check and Treatment Effects

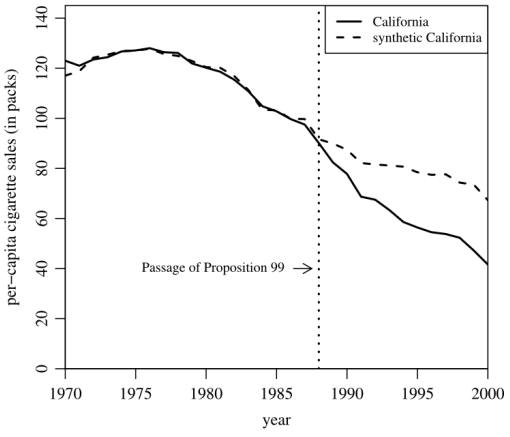
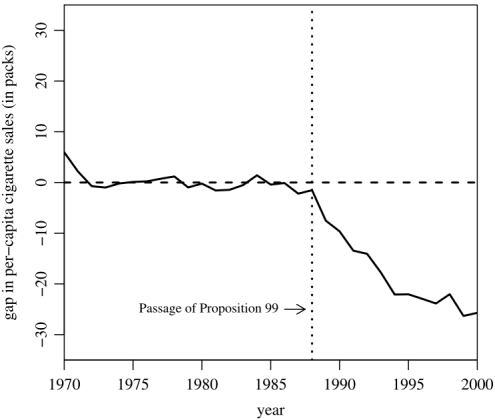


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.



But still some issues

- How sensitive are weights estimates to different covariates?
  - "state-level measures of unemployment, income inequality, poverty, welfare transfers, crime rates, drug related arrest rates, cigarette taxes, population density, and numerous variables to capture the demographic, racial, and social structure of states".
- Can we run a placebo check? Do we detect effects where we know there is a null effect?
  - Put California in the donor pool.
  - Pick a state from the donor pool and pretend that it receives the treatment after T_0
  - Choose w_j following the synthetic control procedure.
  - Compute the treatment effects in the same way.
  - Repeat for all states in donor pool.
  - Compare the ratio of post- to pre-period root mean-square prediction error (RMSPE) across states.
  - This doubles as inference: p-value $=$ rank of California's ratio $/(J+1)$.

Placebo Test

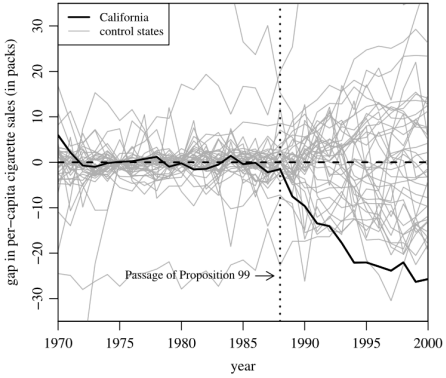


Figure 4. Per-capita cigarette sales gaps in California and placebo gaps in all 38 control states.

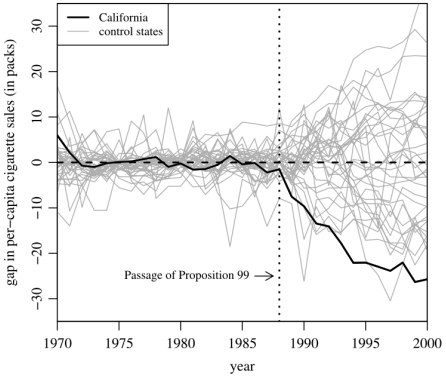


Figure 5. Per-capita cigarette sales gaps in California and placebo gaps in 34 control states (discards states with pre-Proposition 99 MSPE twenty times higher than California's).

More Placebo Tests: How unusual is the CA result?

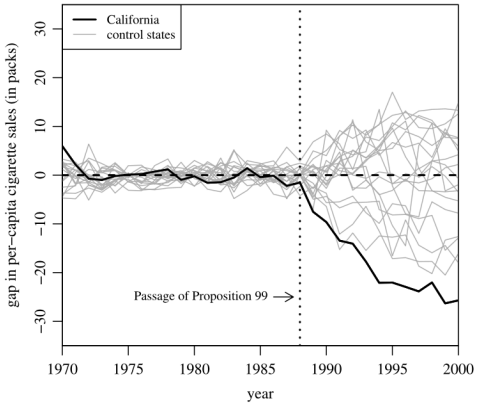


Figure 7. Per-capita cigarette sales gaps in California and placebo gaps in 19 control states (discards states with pre-Proposition 99 MSPE two times higher than California's).

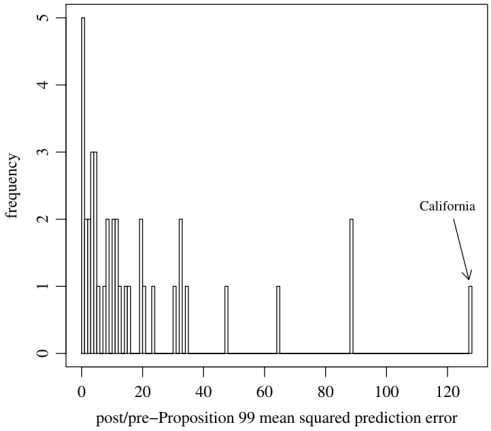


Figure 8. Ratio of post-Proposition 99 MSPE and pre-Proposition 99 MSPE: California and 38 control states.

More Formally

We're going to break this down into three parts:

1. How do we estimate w_j?
2. Conditional on $\widehat{w}_j$'s, how do we estimate the treatment effect?
3. What kinds of models are compatible?

Start with $D_{it} = 1$ IFF $i = 1$ and $t > T_0$:

$$Y_{it} = Y_{it}(0) + \alpha_{it} \cdot D_{it}$$

Estimate $\boldsymbol{\alpha_1} = (\alpha_{1,T_0+1}, \alpha_{1,T_0+2}, \dots, \alpha_{1,T})$ (period by period treatment effect).
For $t > T_0$:

$$\alpha_{1t} = Y_{1t}(1) - Y_{1t}(0) = \underbrace{Y_{1t}}_{\text{observed}} - \underbrace{Y_{1t}(0)}_{\text{counterfactual}}$$

A Starting Point

Consider the model

$$Y_{it}(0) = \delta_t + \theta_t \mathbf{Z}_i + \lambda_t \mathbf{F}_i + u_{it}$$

- δ_t is the time fixed effect
- Z_i are the usual covariates (fixed in i over time) with potentially time varying coefficients.
- λ_t are unobserved common factors: shocks that hit every unit.
- F_i are unobserved i-specific factor loadings: how hard each factor hits unit i. Estimating these is complicated.

This is a generalization of the usual additive $\gamma_t + \gamma_i$ fixed effects model called interactive fixed effects (Bai 2009).

So we define the synthetic observation as:

$$\sum_{j=2}^{J+1} w_j \underbrace{Y_{jt}(0)}_{\text{observed}} = \delta_t + \boldsymbol{\theta}_t \left(\sum_{j=2}^{J+1} w_j \mathbf{z}_j \right) + \lambda_t \left(\sum_{j=2}^{J+1} w_j \mathbf{F}_j \right) + \sum_{j=2}^{J+1} w_j u_{jt}$$

And the difference in untreated observations:

$$\begin{aligned} & Y_{1t}(0) - \sum_{j=2}^{J+1} w_j Y_{jt}(0) \\ &= \boldsymbol{\theta}_t \underbrace{\left(\mathbf{z}_1 - \sum_{j=2}^{J+1} w_j \mathbf{z}_j \right)}_{\text{match this directly}} + \lambda_t \underbrace{\left(\mathbf{F}_1 - \sum_{j=2}^{J+1} w_j \mathbf{F}_j \right)}_{\text{match indirectly: } (Y_{i,1},\ldots,Y_{i,T_0})} + \sum_{j=2}^{J+1} w_j \underbrace{(u_{1t} - u_{jt})}_{\text{Usual } \mathbb{E}[u_{it}|\mathbf{z}_{it},\mathbf{F}_{it}]=0} \end{aligned}$$

Some Algebra

The goal is to match:

$$\mathbf{Z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{Z}_j, \quad \mathbf{F}_1 \approx \sum_{j=2}^{J+1} w_j \mathbf{F}_j$$

- The main result in the Abadie et. al (2010) paper proves that the above equation holds approximately if

$$\left(Y_{1,1}, \dots, Y_{1,T_0}\right) = \sum_j w_j \cdot \left(Y_{j,1}, \dots, Y_{j,T_0}\right)$$

- This frame generalizes the 2WFE model since we can use time dimension to difference them out.
- We cannot difference out the $\lambda_t \mathbf{F}_i$.

The Challenge

Theory depends on matching:

$$\mathbf{z}_1 = \sum_{j=2}^{J+1} w_j \mathbf{z}_j, \quad \left(Y_{1,1}, \dots, Y_{1,T_0}\right) = \sum_j w_j \cdot \left(Y_{j,1}, \dots, Y_{j,T_0}\right)$$

- One problem is the convex hull problem.
  - There may be no $w_2, \dots, w_J$ that satisfy the equations
- Instead choose **w** in a minimum distance sense with some weighting matrix Ω
  - ADH choose Ω (diagonal) so that the resulting synthetic unit minimizes the pre-period MSPE of the *outcome*: a nested optimization (or use cross-validation on a split pre-period).
- Hard to fit crazy out of bounds values: can't match CA population as linear combination of other states!

Other things can go wrong

$$\mathbf{X}_i = \left(\mathbf{Z}_i, Y_{i,1}, \ldots, Y_{i,T_0}\right)$$

$$\arg \min_{w_2,\ldots,w_J} (\mathbf{X}_i - \sum_j w_j \mathbf{X}_j)' \Omega (\mathbf{X}_i - \sum_j w_j \mathbf{X}_j) \quad \text{s.t.} \sum_j w_j = 1$$

- Can get negative weights: what do those mean? (Extrapolation: "short" a donor.)
- We can rule them out with $w_j \geq 0$. For fixed Ω that is a small quadratic program; the hard part is the outer search over Ω, which is non-convex.
- $w_j \geq 0$ and $\sum_j w_j = 1$ put $\mathbf{w}$ on the L_1 ball, the same geometry as the LASSO constraint.
- We shouldn't be surprised by sparse solutions: a handful of donors get all the weight.

Inference with one treated unit

Placebo in space (ADH): treat each donor in turn, $p =$ rank of the treated ratio$/(J+1)$.

- Needs the treated unit to be exchangeable with the donors, which is rarely why it was treated.
- Coarse: with $J = 8$ donors the smallest possible p is $1/9 = 0.11$. You can never reject at 10%.
- Gives a p-value for "no effect", not a confidence interval.

Conformal inference (Chernozhukov, Wüthrich and Zhu 2021, *JASA*): permute over *time* instead of across units.

- Test the sharp null $H_0 : \alpha_{1t} = \theta$ for every $t > T_0$; collect the θ's you cannot reject into a confidence set.
- Needs the residuals of the treated unit around its synthetic control to be stationary and weakly dependent over time. Exact if they are exchangeable.
- The resolution depends on T, not on J.

Conformal inference: the recipe

For a candidate effect θ:

1. Impose the null: $\tilde{Y}_{1t} = Y_{1t} - \theta \cdot \mathbf{1}\{t > T_0\}$.
2. Fit the SC weights on all T periods using $\tilde{Y}_{1t}$. Under H_0 the post-periods are untreated, so they are fair game.
3. Residuals $\hat{u}_t = \tilde{Y}_{1t} - \sum_j \hat{w}_j Y_{jt}$ for $t = 1, \dots, T$.
4. Statistic on the post-period: $S(\hat{u}) = \frac{1}{T - T_0} \sum_{t > T_0} |\hat{u}_t|$.
5. Recompute S on reshuffled residuals; $p(\theta) =$ share at least as large as $S(\hat{u})$.
   - Moving block: all T cyclic shifts of $(\hat{u}_1, \dots, \hat{u}_T)$. Robust to serial correlation; smallest p is $1/T$.
   - iid: random reorderings. Much finer, but assumes no serial correlation.
6. 90% confidence set: $\{\theta : p(\theta) > 0.10\}$.

Intuition: if θ is wrong, the leftover effect sits in the post-period residuals, and they stand out from the pre-period ones.

The soda tax: placebo vs conformal

	40 donors	8 donors
SC estimate (truth -10.7)	-11.2	-10.0
Placebo in space p (floor $1/(J+1)$)	0.024	0.111
Conformal p, moving block (floor $1/T$)	0.083	0.083
90% CI	[-13.7, -0.3]	[-14.2, -0.4]
Conformal p, iid permutations	< 0.001	< 0.001
90% CI	[-13.5, -3.3]	[-12.4, -3.7]

- Placebo p is fine with 40 donors and useless with 8; conformal p does not care how many donors there are.
- With $T = 24$ the moving-block test can't go below $1/24$, and the interval nearly reaches zero. The iid version is tight here because the simulated residuals really are iid. Quarterly sales usually are not.
- augsynth: $\mathrm{summary}(\mathrm{fit},\ \mathrm{inf_type} = \text{"conformal"})$ gives per-period intervals (iid by default). $\mathrm{conformal_sims.R}$ does it by hand in about ten lines.

In practice

- Default: augsynth (Ben-Michael, Feller, Rothstein; install from GitHub, actively maintained).
  - progfunc = "None" is classic SC; "Ridge" adds the augmented-SC bias correction.
  - Conformal inference built in; multisynth handles staggered adoption.
- tidysynth: easiest placebo machinery (grab_significance gives the RMSPE-ratio p-value).
- synthdid: SC, DiD, and SDID side by side (next section); no longer actively maintained.
- The original Synth (which tidysynth wraps) has optimization issues:
  - When no weights fit exactly and the non-negativity constraints bind, the outer search over Ω is badly behaved
  - It can produce sub-optimal solutions (and slowly). MSCMT fixes this.

A more unified view

We can ask – what are the implied regression weights for linear regression? What is $\mathbb{E}[Y_{1,t}(0) \mid \mathbf{X}]$?

- Get coefficients using just donor pool: $\widehat{\mathbf{B}} = \left(\overline{\mathbf{X}}_0 \overline{\mathbf{X}}_0'\right)^{-1} \overline{\mathbf{X}}_0 \mathbf{Y}_0'$
- Predict $Y_{1,t}(0) = \widehat{\mathbf{B}}' \overline{\mathbf{X}}_1$.
- Same as synthetic control but with weights: $\mathbf{W}^{reg} = \overline{\mathbf{X}}_0' \left(\overline{\mathbf{X}}_0 \overline{\mathbf{X}}_0'\right)^{-1} \overline{\mathbf{X}}_1$ (Projection matrix).

Why? Remember, everything is a kernel our prediction of $\mathbb{E}[Y \mid X]$ is always some weighted average of other Y_j's.

Additional Applications

- What is word of mouth effect of Superbowl ads? (Lovett, Peres, Xu QME 2019)
  - Construct synthetic versions of brands that don't buy Superbowl ads
- What is effect of soda tax on consumption in Berkeley, CA ? (Bollinger Sexton WP 2019).
  - Construct synthetic drugstores, supermarkets, and convenience stores using two donor pools: inside CA and outside CA.
- What are economic effects of German reunification?
  - Now we need a counterfactual "West Germany" (!)
    (Abadie, Diamond, and Hainmueller 2015)

TABLE 1 Synthetic and Regression Weights for West Germany

Country	Synthetic Control Weight	Regression Weight	Country	Synthetic Control Weight	Regression Weight
Australia	0	0.12	Netherlands	0.09	0.14
Austria	0.42	0.26	New Zealand	0	0.12
Belgium	0	0	Norway	0	0.04
Denmark	0	0.08	Portugal	0	−0.08
France	0	0.04	Spain	0	−0.01
Greece	0	−0.09	Switzerland	0.11	0.05
Italy	0	−0.05	United Kingdom	0	0.06
Japan	0.16	0.19	United States	0.22	0.13

Notes: The synthetic weight is the country weight assigned by the synthetic control method. The regression weight is the weight assigned by linear regression. See text for details.

TABLE 2 Economic Growth Predictor Means before German Reunification

	West Germany	Synthetic West Germany	OECD Sample
GDP per capita	15808.9	15802.2	8021.1
Trade openness	56.8	56.9	31.9
Inflation rate	2.6	3.5	7.4
Industry share	34.5	34.4	34.2
Schooling	55.5	55.2	44.1
Investment rate	27.0	27.0	25.9

Notes: GDP per capita, inflation rate, trade openness, and industry share are averaged for the 1981–90 period. Investment rate and schooling are averaged for the 1980–85 period. The last column reports a population-weighted average for the 16 OECD countries in the donor pool.

FIGURE 1 Trends in per Capita GDP: West Germany versus Rest of the OECD Sample

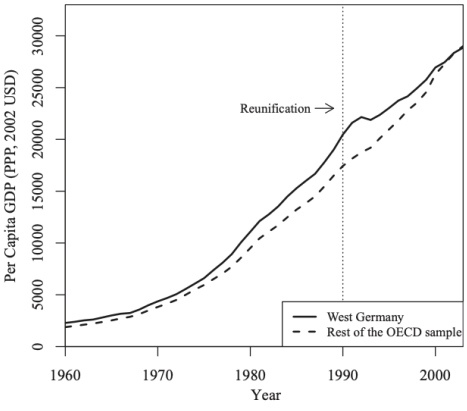


FIGURE 2 Trends in per Capita GDP: West Germany versus Synthetic West Germany

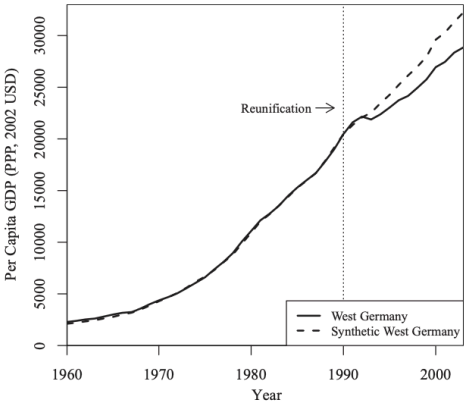


FIGURE 3 Per Capita GDP Gap between West Germany and Synthetic West Germany

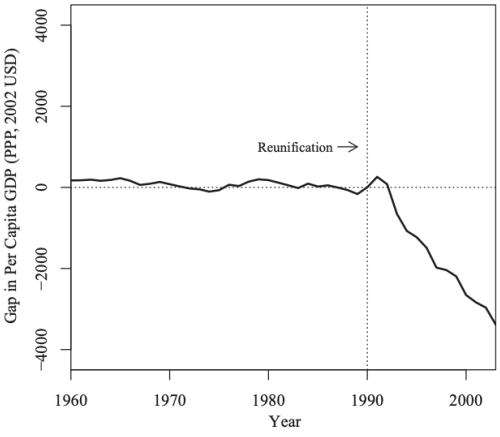
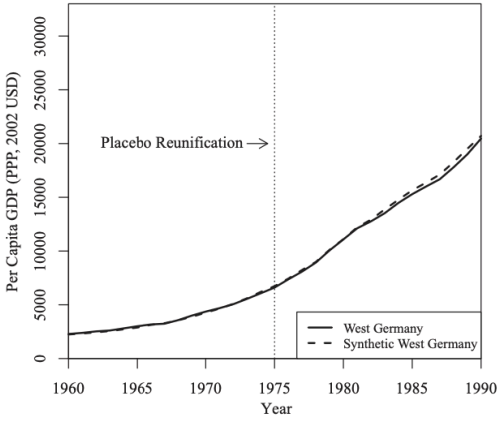
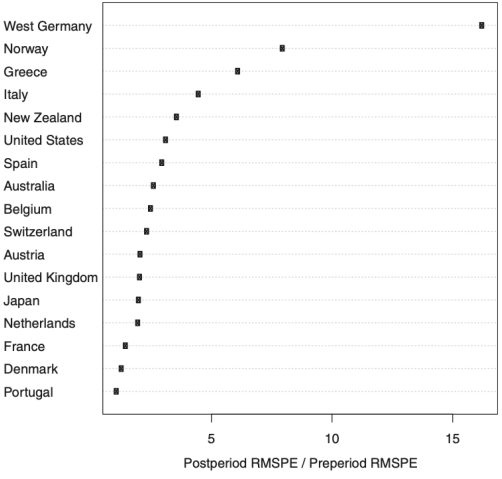


FIGURE 4 Placebo Reunification 1975–Trends in per Capita GDP: West Germany versus Synthetic West Germany



Treatment Effects and Placebo Timing

FIGURE 5 Ratio of Postreunification RMSPE to Prereunification RMSPE: West Germany and Control Countries



Synthetic difference-in-differences

Three estimators, one formula

Arkhangelsky, Athey, Hirshberg, Imbens and Wager (2021) write all three as weighted two-way fixed-effects regressions:

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min \sum_j \sum_t \left(Y_{jt} - \mu - \alpha_j - \beta_t - \tau W_{jt}\right)^2 \omega_j \, \lambda_t$$

	unit weights ω_j	time weights λ_t	unit fixed effects α_j?
DiD	equal	equal	yes
Synthetic control	fit pre-period, no intercept	equal	no
Synthetic DiD	fit pre-period *trends*, intercept allowed	fit pre-to-post, intercept allowed	yes

DiD trusts levels (fixed effects) but not the control group's composition; SC trusts composition (weights) but insists on matching levels. SDID reweights donors to track the treated unit's *trend*, absorbs the level gap with α_j, reweights pre-periods to resemble the post-period, then differences.

How the weights are chosen

- Unit weights $\hat{\omega}$: nonnegative, sum to one, with a free intercept ω_0,

$$\min_{\omega_0,\omega} \sum_{t \le T_0} \left(\omega_0 + \sum_j \omega_j Y_{jt} - Y_{1t} \right)^2 + \zeta^2 T_0 \|\omega\|^2.$$

  The intercept means the synthetic unit only has to *move* like the treated unit; the ridge penalty ζ spreads weight across donors.
- Time weights $\hat{\lambda}$: nonnegative, sum to one, with an intercept, chosen so the weighted pre-period average of each donor predicts its post-period average. Pre-periods that look like the post-period count more; the distant past counts less.
- Then run the weighted two-way FE regression. $\hat{\tau}$ is a DiD in which the control group and the pre-period have both been reweighted to resemble the treated unit and the post-period.
- Inference: placebo, jackknife over units, or bootstrap; the placebo variance works well with few treated units. `synthdid` in R implements all of it.

The soda tax again: DiD, SC, and SDID on simulated cities

	DiD (all donors)	Synthetic control	Synthetic DiD	Truth
Trends differ, levels similar (treated inside the hull)	-9.4	-11.2	-11.1	-10.7
Trends differ *and* treated level above every donor	-11.1	-8.9	-12.0	-12.2
Treated trend outside the hull (loading 1.45)	-6.4	-6.7	-9.7	-11.3
Bias, 60 draws of the level-shift design	1.31	5.47	0.88	
RMSE, 60 draws of the level-shift design	1.52	6.19	1.28	

One treated city, 40 donors, 24 quarters, tax from quarter 17. When the trend can be matched and the level is ordinary, SC and SDID both beat DiD. When the treated city sits *above every donor* in level, SC cannot reproduce it (no intercept) and is off by about a quarter; DiD handles the level but not the trend; SDID handles both, with a sixth of SC's bias over repeated draws. When the *trend* is outside the hull, nothing recovers the truth, and the pre-period fit says so (SC's pre-period RMSPE triples).

Which one, and when

- Many treated units, parallel-ish trends, levels differ: DiD, with the Week 9 cautions about staggered timing.
- One or a few treated units, a donor that can be built by convex combination: SC. Report the pre-period fit and the placebo distribution; it is the design's only credibility.
- One or a few treated units whose level is unusual but whose trend is not: SDID. The intercept is doing the work; it is the same idea as running DiD after picking a better control group.
- Treated unit is extreme in trend: none of them. Say so.
- All three are weighted DiDs. The question is always the same: which units and which periods make an honest counterfactual, and what did you have to assume to believe it.

Thanks